

AUDEN REID-MCLAUGHLIN

PhD Candidate in Geophysics
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Google Scholar

California Institute of Technology
Expected Graduation: Summer 2027
Advisors: Zhongwen Zhan and Nadia Lapusta

DISCIPLINARY FIELDS

Geophysics; Cryoseismology; Distributed Acoustic Sensing (DAS); Active-Source Seismology; Seismic Imaging; Fault Mechanics; Computational Geophysics

EDUCATIONAL BACKGROUND

PhD, California Institute of Technology, Geophysics Expected Summer 2027
Advisors: Zhongwen Zhan and Nadia Lapusta
B.S., Cornell University, Earth and Atmospheric Sciences 2022
Magna Cum Laude; Minor in English; GPA: 3.90

RESEARCH AND FIELD EXPERIENCE

Grounding Zone Long-term Acoustic Sensing of Structure (GLASS), Antarctica 2025–present
California Institute of Technology

- PhD research investigating grounding-zone structure, basal conditions, and tidal variability using active-source DAS, dense nodal seismology, radar, and geodetic observations.
- Helped plan and execute a 12-day field campaign at Constellation Inlet, Antarctica, with responsibility for two DAS systems, approximately 500 STRYDE geophones, and 20 three-component SmartSolo stations.
- Lead development and application of the seismic processing, imaging, and interpretation workflows for the project, spanning approximately 3 TB of DAS and 500 GB of nodal observations.
- Co-authored the successful Brinson Foundation proposal supporting the 2025–2027 project.

Post-Eaton Fire Rapid-Response Debris-Flow Monitoring, California 2025
California Institute of Technology

- Led DAS field deployment and analysis for rapid-response monitoring of post-fire debris flows in Bailey Canyon following the 2025 Eaton Fire.
- Developed methods to detect and characterize debris flows from dense fiber-optic observations, including measurements of flow-front velocity and triggering intensity.

South Pole Distributed Acoustic Sensing, Antarctica 2025
California Institute of Technology

- Led active-source analysis and seismic modeling of an 8,225-channel DAS experiment to investigate englacial structure and conditions at the base of the Antarctic ice sheet.
- Combined observations and synthetic modeling to constrain ice properties, attenuation, basal temperature, and subglacial sedimentary conditions.

PUBLICATIONS AND MANUSCRIPTS

Peer-Reviewed Publications

- Y. Yang, Z. Zhan, M. Karrenbach, **A. Reid-McLaughlin**, E. Biondi, D. A. Wiens, and R. C. Aster (2024). *Characterizing South Pole Firn Structure With Fiber Optic Sensing*. Geophysical Research Letters, 51, e2024GL109183. doi:10.1029/2024GL109183.

Manuscripts in Review

- **A. Reid-McLaughlin**, Z. Zhan, M. Karrenbach, Y. Yang, E. Biondi, Q. Zhai, D. A. Wiens, and R. C. Aster. *Constraining Temperature and Sedimentary Conditions under the South Pole with Distributed Acoustic*

Sensing. Manuscript in review.

- X. Wei, E. Geyman, Z. Chen, **A. Reid-McLaughlin**, M. Lamb, and Z. Zhan. *Triggering and evolution of debris flows following the 2025 Los Angeles Fires captured by seismometers*. Manuscript in review.
- T. Liu, **A. Reid-McLaughlin**, and Z. Zhan. *Englacial reflector at the South Pole revealed by Distributed Acoustic Sensing*. Manuscript in review.

Manuscripts in Preparation

- **A. Reid-McLaughlin**, N. Lapusta, and L. Dal Zilio. *Processing and Parameter Dependence of Slow-Slip Scaling in a Rate-and-State Framework*. Manuscript in preparation.
- **A. Reid-McLaughlin**, Z. Chen, X. Wei, E. Biondi, and Z. Zhan. *Rapid Response DAS Deployment for in-situ monitoring of post-Eaton Fire Debris Flows*. Manuscript in preparation.

TEACHING EXPERIENCE

California Institute of Technology

- Field Geophysics – Teaching Assistant (Spring 2024)
- Plate Tectonics – Teaching Assistant (Fall 2024)
- Fracture and Frictional Faulting – Teaching Assistant (Spring 2025)
- Rock Mechanics – Teaching Assistant (Winter 2026)

HONORS AND AWARDS

Resnick Scholar for Sustainability, California Institute of Technology

Current

Magna Cum Laude, Cornell University

2022

Dean's List, Cornell University

All semesters

FUNDED RESEARCH

Grounding Zone Long-term Acoustic Sensing of Structure (GLASS)

2025–2027

Brinson Foundation-funded research project. Co-author of funded project proposal.

TECHNICAL EXPERTISE

Seismology and Geophysical Imaging: Distributed acoustic sensing (DAS); active-source seismology; seismic imaging; array seismology; seismic waveform and amplitude analysis.

Computational Geophysics: Numerical wave-propagation modeling; fault and earthquake mechanics; rate-and-state friction modeling; inverse and quantitative methods.

Scientific Computing: Python, MATLAB, and Fortran; high-performance and parallel computing; large-scale scientific data analysis; Linux/Unix.

Field Geophysics: Large-scale seismic and fiber-optic deployments; active-source experiments; dense nodal arrays; multi-instrument geophysical surveys; polar and rapid-response field operations.

OUTREACH AND SERVICE

Earthquake Fellows Program, California Institute of Technology

3 summers

- Instructor and mentor in a four-week summer program teaching seismology to high-school students.

Union Steward, GPS, California Institute of Technology

Current

Seismology Reading Group, Organizer

2024–Present

- Organized weekly/monthly reading groups for Cryoseismology and Environmental Geophysics.